Outcome-Aligned On-Policy Distillation with Rank-Calibrated Token Supervision

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Abstract

On-policy distillation (OPD) transfers token-level preferences from a stronger teacher to samples generated by the student itself, making it attractive for improving reasoning models without leaving the student distribution. However, dense teacher rewards are not uniformly reliable: a teacher can assign local token preferences that conflict with whether the whole sampled trajectory is ultimately correct. We propose Outcome-Aligned Logit OPD (OAL-OPD), which treats outcome correctness as a trajectory-level calibration signal for token-level distillation. OAL-OPD uses rank-calibrated token supervision: it preserves full teacher supervision for outcome-aligned token candidates while retaining a small, bounded rank-based learning signal for anti-aligned candidates. This design reduces the influence of outcome-conflicting dense rewards without discarding all information from uncertain local decisions, yielding a controlled mechanism for studying and improving token-level credit assignment in OPD.

Introduction

Large language models have shown strong gains from reinforcement learning and distillation on mathematical reasoning tasks. A particularly direct approach is on-policy distillation (OPD), where a student first samples its own responses and then receives dense token-level supervision from a stronger teacher on those on-policy trajectories. Compared with offline imitation, OPD avoids training solely on teacher-generated data and can adapt the teacher signal to the actual states visited by the student. In reasoning tasks, this property is valuable because small models often visit different intermediate states from their teachers.

Despite this advantage, OPD relies on a strong assumption: the teacher's local token preference is a reliable training signal for every token on the student trajectory. This assumption can fail. A sampled response is evaluated only after the whole solution is produced, while the teacher provides dense token-level differences at each intermediate step. As a result, some token-level teacher rewards may be misaligned with the final outcome. For example, in a correct response, the teacher may locally prefer reducing a token candidate that is nevertheless part of a successful trajectory; conversely, in an incorrect response, the teacher may locally increase candidates that reinforce an ultimately wrong solution path. Such conflicts make token-level credit assignment a central bottleneck for OPD.

The final correctness of a reasoning trajectory provides a natural global check on these local teacher preferences. Although outcome feedback is sparse, it directly tells us whether the overall reasoning path succeeds. We therefore use the outcome not as an additional reinforcement learning loss, but as a calibration signal for deciding which token-level teacher preferences are consistent with the result. If a trajectory is correct, candidate tokens that the teacher prefers more than the student are more likely to support successful reasoning. If a trajectory is incorrect, the aligned direction is reversed: teacher preferences that reduce the likelihood of the sampled behavior are more consistent with the failed outcome. This converts trajectory-level correctness into a reliability criterion for dense OPD supervision, yielding a simple but informative partition of token candidates into outcome-aligned and outcome-anti-aligned groups.

We propose Outcome-Aligned Logit OPD (OAL-OPD), an OPD objective that uses this partition to rank-calibrate dense teacher rewards. OAL-OPD keeps aligned candidates at full strength and assigns anti-aligned candidates a small rank-based weight. This makes OAL-OPD a conservative modification of OPD: it preserves the main dense teacher signal when it is outcome-consistent, while using a bounded auxiliary signal to avoid overcommitting to a binary token decision under noisy local and global credit assignment.

Our method is designed to answer two questions. First, can final correctness provide a useful calibration signal for token-level distillation? Second, when token-level teacher rewards conflict with the final outcome, how much influence should those tokens retain? OAL-OPD addresses the second question with beta-capped rank weighting over anti-aligned candidates. We further conduct a detailed experimental analysis of aligned and anti-aligned token candidates, explaining their respective roles during training.

The contributions of this work are:

- We identify outcome-token mismatch as a practical failure mode of dense OPD rewards in long-horizon reasoning.

- We formulate outcome-aligned token supervision using the sign of teacher-student log-probability differences together with trajectory-level correctness.
- We introduce rank-calibrated OAL-OPD, which preserves aligned teacher supervision while softly down-weighting anti-aligned candidates using a bounded rank score.
- We provide a logging and analysis framework for studying how aligned and anti-aligned token candidates affect reasoning performance.

Method

On-Policy Distillation

Let π_S denote the student policy and π_T denote the teacher policy. Given a prompt, the student samples a response $x_{1:T}$ from π_S. OPD computes a dense teacher reward for candidate token $a_{t,k}$ at position t using the teacher-student log-probability difference:

$$\Delta_{t,k} = \log \pi_T(a_{t,k} \mid x_{<t}) - \log \pi_S(a_{t,k} \mid x_{<t}). \tag{1}$$

The candidate set is usually restricted to a top-K subset for efficiency. A standard OPD objective directly uses $\Delta_{t,k}$ as a dense advantage-like signal. This treats all token-level teacher preferences as equally reliable, regardless of whether the sampled response is correct.

Outcome-Aligned Token Partition

Let $r \in \{0, 1\}$ be the trajectory-level outcome, where $r = 1$ means the sampled response is correct and $r = 0$ means it is incorrect. We map this binary outcome to a signed label:

$$y = 2r - 1. \tag{2}$$

For each candidate token, we define an outcome-aligned score:

$$s_{t,k} = y\Delta_{t,k}. \tag{3}$$

When $s_{t,k} > 0$, the teacher-student preference agrees with the final outcome. In a correct response, this means the teacher assigns higher probability than the student to the candidate. In an incorrect response, this means the teacher assigns lower probability than the student, discouraging behavior observed in a failed trajectory. We use a margin $m \geq 0$ to avoid ambiguous cases:

$$\mathcal{A} = \{(t, k) : s_{t,k} > m\}, \quad \mathcal{B} = \{(t, k) : s_{t,k} < -m\}, \tag{4}$$

where $\mathcal{A}$ contains outcome-aligned candidates and $\mathcal{B}$ contains outcome-anti-aligned candidates.

Rank-Calibrated OAL-OPD

OAL-OPD keeps the main outcome-aligned supervision intact while softly retaining a bounded signal from anti-aligned candidates. For each response, we rank only candidates in $\mathcal{B}$ according to their outcome-aligned score $s_{t,k}$. Let $q_{t,k} \in [0, 1]$ be the normalized rank of an anti-aligned candidate within that response. Candidates with relatively larger $s_{t,k}$ are closer to the decision boundary and receive larger rank values. We then define

$$w^{\text{rank}}_{t,k} = \begin{cases} 1, & s_{t,k} > m, \\ \beta q_{t,k}, & s_{t,k} < -m, \\ 0, & |s_{t,k}| \leq m. \end{cases} \tag{5}$$

where $\beta \in [0, 1]$ controls the maximum retained weight for anti-aligned candidates. The final dense advantage is

$$A_{t,k} = w^{\text{rank}}_{t,k} \Delta_{t,k}. \tag{6}$$

Unlike a rank scheme that rescales all candidates, OAL-OPD never down-weights outcome-aligned candidates. The ranking is applied only to the anti-aligned region, so β has a clear meaning: it is the largest possible supervision strength assigned to candidates whose local teacher preference conflicts with the final outcome.